

FSK176: Homework Problem Numbers / Huiswerk Probleem Nommers

2017

2 Week: 24 July / 24 Julie 2017

Chapter 2 Motion in 1 dimension / Hoofstuk 2 Beweging in 1 dimensie

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P2.5 (a) 2.30 m/s (b) 16.1 m/s (c) 11.5 m/s

P2.7 (a) -2.4 m/s (b) ≈ -3.8 m/s (c) 4.0 s

P2.21 (a) 2.00 m (b) -3.00 m/s (c) -2.00 m/s²

P2.25 (a) 4.98×10^{-9} s (b) 1.20×10^{15} m/s²

P2.30 (a) 20.0 s (b) 1000 m is needed to land (c) The plane can not land.

P2.56 (a) $|v_p| = v_i + gt$ (b) $d = \frac{1}{2}gt^2$ (c) $v_p = |v_i - gt|$ and $d = \frac{1}{2}gt^2$

P2.63 (a) 3.00 s (b) -15.3 m/s (c) $v_{1f} = -31.4$ m/s and $v_{2f} = -34.8$ m/s

P2.72 (a) 41.0 s (b) 1.73 km (c) -184 m/s

P2.75 (a) $v_B = \left(\frac{1}{\tan \theta}\right) v$ (b) Explanation